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Research on key performance indicator (KPI) of business process

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***Abstract*—The execution of business process is a management process for the integration of goals, systems, processes, information and persons. While most Chinese enterprises are faced with the main problem which is business process management that cannot be implemented effectively, thus it can not achieve the desired target value. So key performance indicator (KPI) evaluation system is added in the enterprise informatization framework that can change the dynamic structure of business process quickly, thus make it meet rapid changes in market demand.**

Keywords-KPI; business; process; KPI performance evaluation

I. INTRODUCTION

At present, many enterprises have established key performance indicator (KPI) evaluation system relatively perfect, but their management mode was limited to manual management. Due to the increasing of the fierce market competition and the scale of enterprises, the methods of manual computation and management have been unable to cope with the large number of statistical data and rapid market changes. Therefore, business process management of information system's demand becomes more and more urgent. With the development of the information technology, business process management of information system has become feasible. The application of this system can enhance enterprise's creation ability effectively, and it also could make business management system more perfectly. However, this kind of management system and other management systems has existence independence, they can not share data effectively, and more seriously is that it lacks of sufficient information for changes of market and the business process, and it can not adjustment for the real-time of key performance indicators [1].

Therefore, from the view of the above problems, enterprises require the combination of the original independent KPI performance appraisal system informatization and business process thought and platform results, design and execute KPI performance evaluation system. KPI performance evaluation system is based on business process driven, also according to business process change and the dynamic change of KPI performance assessment, in order to meet the fast changing market demand.

II. BUSINESS PROCESS MANAGEMENT (BPM)

Citing the Business Process Management (BPM) Alliance on business process definition: business process is an ordered set which enterprises accomplish a goal or a task, and carry out a series of logically related activities. It basically solves the problem which is "in order to achieve a desired business goal or make this goal become true, make multiple participants finish activities or tasks automatically according to the predefined rules of the transmission of documents and information." [2]. In brief, business process is a series of interlocking and automatic business activities or tasks. A business process includes a set of tasks or activities and their mutual relation of sequences, and it also includes processes and tasks or activities in the initiation and termination conditions, as well as it has descriptions on each task or activity. Obviously, BPM not only covers the traditional "workflow" process transfer, process monitoring category, but also it is a breakthrough of the traditional "workflow" technical bottleneck. The introduction of BPM is an epoch-making leap for the workflow technology and the concept of the enterprise management [3].

However, the current business process management software still reflect the data through reports or data flow diagrams. That is to say, the performance appraisal systems and business process management systems have existence independence. So it is inefficient and inconvenient for business process assessment.

III. INTERGRATED KPI INTO BUSINESS PROCESS MANAGEMENT

Key performance indicator (KPI) is a key driving factor which implements the strategic objectives of enterprises and it is an evaluation index of the core events, it formulates around the enterprise strategies, and it is a form of expression for the quantified strategy [4]. Key performance indicators are used to measure staffs' work performance indexes, so it is the important part of the performance plans [5].

In the business process management, the performance management is a very important link. The method of key performance indicator was implemented in the performance evaluation by various types of enterprises. This should be attributed to two striking features of KPI:

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1) KPI emphasizes on the performance indicators that must be configured with the organizational development strategy;

2) KPI concerns the problems which the organization is most in need of attention and urgent to resolve in specific period [6].

In today's business process management software, the data is mainly embodied from two aspects:

1) Using the report form;

2) Using the data flow diagram.

These two methods do not reflect the key performance indicators. In the business process management, the calculation ability of the key performance indicators is crucial. Without these abilities, it can't optimize the business processes intelligently or in response to strategic events to create a valid new business process. Therefore, added the method of key performance indicator to the business process model had optimized the function of business process and had solved the enterprise's urgent need in the business process management.

IV. THE APPLICATION ANALYSIS

Take the printing business process as an example, its organization structure is shown in Figure1:

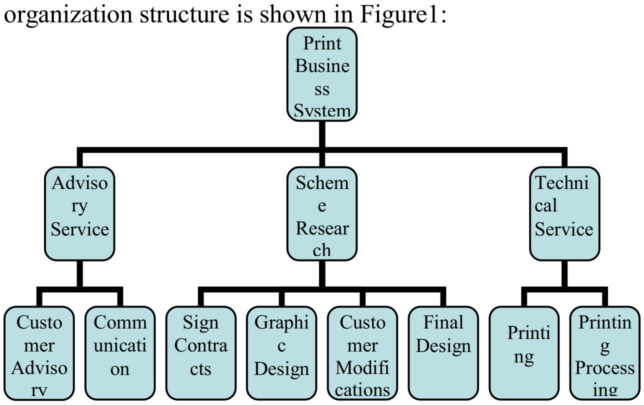

Figure 1. Organization Structure

In the printing business process, the related KPI indicators are divided into three categories: the quantity, quality and efficiency. KPI indicators and their definition are shown in Table 1:

In the printing business system, advisory service includes two steps:

Step1: Customer Advisory online and by hotline;

Step2: Professional staffs went to the door of communication.

As is shown in Figure 2:

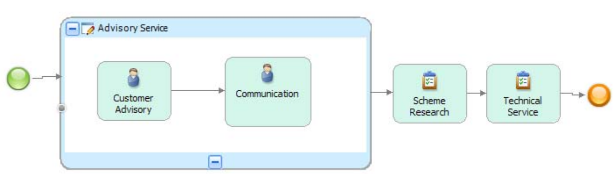

Figure 2. Printing Business Process

Take step1 as an example, according to efficiency, it mainly divides into three circumstances. To define parameters of KPI firstly. As is shown in Figure 3:

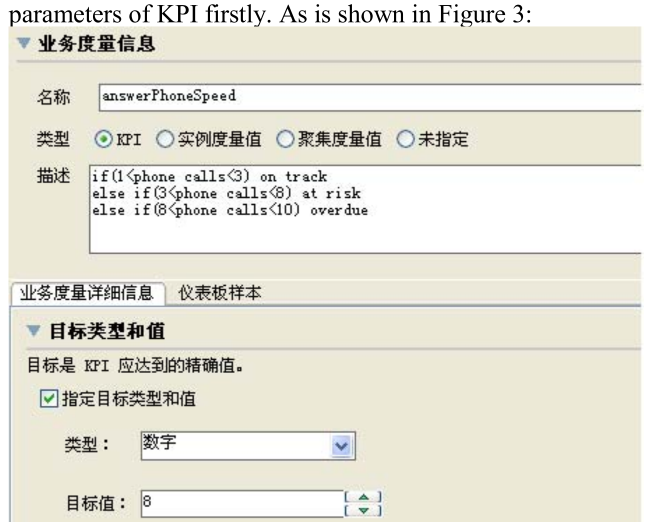

Figure 3. Define parameters of KPI

As is shown in Figure 3, set the KPI target type of the preliminary work time is "number", set the value is "eight". Eight is the limit value of the "answerPhoneSpeed".

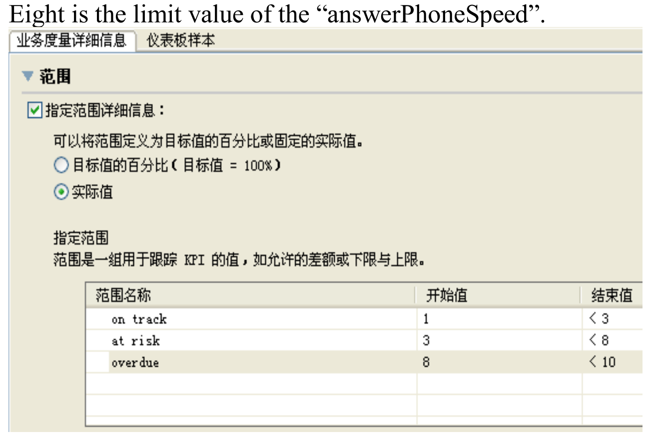

Figure 4. Set the range of KPI

Figure 4 shows the range of KPI, the assessment is the actual value of the completion time. Since the step1 for a period of 3, so set the name of range from 1 to 3 as "on track", set the name of range from 3 to 8 as "at risk", set the name of range from 8 to 10 as "overdue".

The analysis of the result is a pie chart, in the picture we can clearly see the proportions of "on track", "at risk" and "overdue". As is shown in Figure 5:

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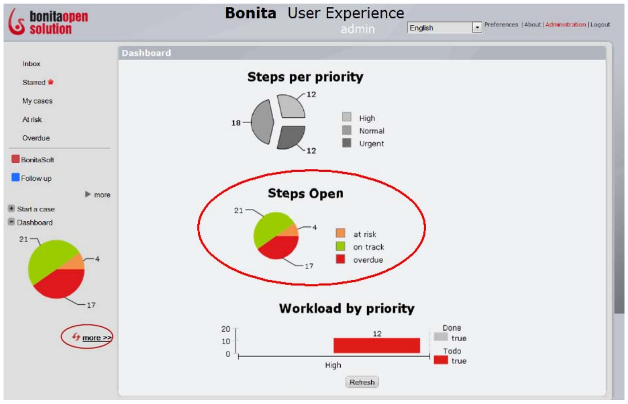

Figure 5. The analysis of the result

Through this figure, not only can managers appraise of staffs' performances, but also can they adjust KPI index timely. It makes the index more beneficial for the completion of the tasks or activities.

We can find the advantages of defining KPI in the business process through the analysis of the result:

1) It can improve the visualization of the performance appraisal;

2) It can increase the contrastive of assessment result;

3) It can be applied to the results of the assessment, and feedback it to the KPI performance evaluation system, thus optimize the KPI performance appraisal system.

V. CONCLUSION

From the view of the enterprise business process management, how to make strategies which are concerned by senior management to be throughout the entire enterprise organization, personnel and business execution process, the lack of an effective operational management process system to ensure the implementation of the process. This paper is based on the KPI operation management system, it emphasizes on the strategic planning layer and the operation layer oriented toward the implementation of the operation management process. It constructs the closed loop combined with the KPI system and the adaptive enterprise operation management system, improving the executive power of the enterprises, in order to promote the implementation of the system strategy.

And domestic definition of the KPI in the business process is in the primary stage, it needs to be further studied, and it is a significant contribution to improve the business process management.

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TABLE I. KPI INDICATORS AND THEIR DEFINITION

KPI Indicator Type	KPI Indicator Name	Indicator Definition / Formula
Quantity	answerPhoneNumber	answerPhoneNumber / allPhoneNumber *100%
	replyEmailNumber	replyEmailNumber /allEmailNumber*100%
	communicationNumber	communicationNumber/allcommunicationNumber *100%
Quality	clientSatisfaction1	clientSatisfaction1 >=85，excellent
		70< clientSatisfaction1 <85，pass
		clientSatisfaction1 <70，fail
	clientSatisfaction2	clientSatisfaction2>85，excellent
		70< clientSatisfaction2<85，pass
		clientSatisfaction2<70，fail

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	fileSatisfaction	fileSatisfaction >85，excellent
		70< fileSatisfaction <85，pass
		fileSatisfaction <70，fail
	printQuality	qualified number/all number*100%
Efficiency	answerPhoneSpeed	Phone calls<3，excellent
		3< Phone calls <8，pass
		Phone calls >8，fail
	communicationSpeed	visitTime<2 days，excellent
		2 days< visitTime <3days，pass
		visitTime >3 days，fail
	finishFileSpeed	speedTime/planTime*100%
	printSpeed	printTime/planTime*100%
	replyEmailSpeed	replyTime<5 hours，excellent
		5 hours< replyTime <1day，pass
		replyTime >1 day，fail

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